Credit Name: CSE 2110 Procedural Programming 1
Assignment Name: Chapter 5 Mastery Exercise 7 (PerfectInteger)

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem
How did you approach understanding the challenge?
Working through the IPO chart and carefully reading the criteria in the textbook before I started coding was how I figured out how I wanted to structure the code. An input would not be required, just the main method and a helper boolean method to determine whether or not a number was a perfect integer. The perfect integers would then be displayed for the user.
Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?
I understood the problem and the necessary steps, but choosing the correct tools and getting the details right was tricky. I simplified it by breaking the challenges into smaller steps and using available resources for support when I got stuck.

Planning the Solution
Did you create a plan or break the problem into smaller steps before coding?
Yes, the code was tricky for me so I tested small steps individually. Such as the loops, the calculations, and the methods.
How did you decide on the tools, data structures, or algorithms to use?
I used integer variables because the numbers had to be positive integers from 1 - 100. I used for loops because I already knew the exact range of numbers I had to test. If statements helped allow the code to only run a step if certain criteria was met.

Implementation
Did you write the code in small pieces or attempt the entire solution at once?
While I was coding I would test small sections such as calculations, the for loops, if statments... Just to make sure all the parts were working properly for when I put it all together.
How did you test your solution along the way to make sure it was working?
I would just run a few lines of test code (or the old code, plus the new lines I just added) to see if it worked. This helped me catch errors early and be able to fix them without entirely rewriting my code.

Overcoming Challenges
What part of the problem was the most difficult for you?
Getting all the details perfect, and making sure all the parts worked together properly was difficult, but there wasn't really a specific problem that was more challenging.
How did you handle moments when you felt stuck or unsure of what to do next?
I asked some friends what they would do or I would refer to the latest chapters in the textbook.

Learning
Was there anything you learned that you think will help you with future challenges?
In this code I learned some practicle applications of boolean methods as well as the usefulness of an if statement nested in a for loop, and how to use the loops to test a number different integers.